Prateek Sinha

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Summary

AI/ML enthusiast with hands-on experience in building LLMs, Retrieval-Augmented Generation (RAG) systems, and reinforcement learning environments. Proficient in Data Structures & Algorithms in Python. Skilled in fine-tuning transformer models, developing end-to-end AI applications, and deploying full-stack solutions. Strong hackathon performer focused on real-world AI systems.

Education

SRM Institute of Science and Technology — Kattankulathur, Tamil Nadu
B.Tech in Computer Science and Engineering — *2023 – 2027*
- CGPA: 8.36

Projects

RAG Chatbot System | *FastAPI, ChromaDB, Sentence-Transformers, Next.js* — Oct 2024
- Built end-to-end RAG system for document QA achieving ~85% retrieval relevance and ~40% fewer hallucinations vs. baseline
- Designed ingestion pipeline with chunking, embedding, and ChromaDB vector search
- Developed Next.js frontend with FastAPI backend; real-time query handling under 2s response time

LawGPT | *Python, PyTorch, Transformers* — May 2025
- Fine-tuned GPT-2 on 50K+ Indian legal QA pairs (IPC, CPC) for domain-specific question answering
- Instruction tuning pipeline via Hugging Face Transformers; 30% faster training with gradient checkpointing
- Built inference pipeline with context-aware legal responses at avg. under 3s latency on CPU

APEX AI Survival Arena | *PyTorch, Pygame, NumPy* — Sept 2025
- Built multi-agent RL system using DQN with 8+ agents; achieved 3x survival rate over random baseline in 500 episodes
- Designed 12×12 grid environment with resource management, seasonal dynamics, and 11-action/25-dim state space
- Implemented experience replay and target network stabilization; reduced training loss ~60% vs. vanilla Q-learning

LLM Adaptation Study | *Python, PyTorch, Transformers, PEFT* — May 2026
- Compared Prompt Engineering, LoRA, and Full Fine-Tuning across transformer-based NLP tasks
- Benchmarked model quality, training cost, and resource utilization using parameter-efficient fine-tuning techniques
- Analyzed trade-offs between performance and deployment efficiency for LLM adaptation strategies

Racecraft AI | *React.js, TypeScript, Python, Groq LLM, SQLite, FastF1* — Jun 2026
- Built an agentic AI platform for Formula 1 analytics using Retrieval, Memory, Prediction, and Reasoning agents
- Developed a Circuit Memory Engine and historical analytics pipeline for drivers, constructors, and race performance
- Implemented explainable race prediction models and an AI assistant for F1 insights and forecasting

Achievements

Top 25 AIR – HackHazard 2025 (Monad Track) for QuestMint

Top 100 AIR – HackHazard 2025 (Overall)

Certifications

GitHub Copilot – Microsoft (Apr 2026) | Verify

Machine Learning with Python (ML0101EN) – IBM / Cognitive Class (Oct 2025) | Verify

Artificial Intelligence Fundamentals – IBM (Jul 2026) | Verify

Career Essentials in Generative AI – Microsoft and LinkedIn (Jun 2026) | Verify

Technology Job Simulation – Deloitte (Jun 2026) | Verify

Technical Skills

Languages: Python (DSA, AI/ML – Intermediate), SQL, HTML, CSS, JavaScript
Tools and Framework: Streamlit, Flask, Fast API, AutoCAD, Git
AI/ML: PyTorch, Pygame, Transformers, TensorFlow, LangChain, ChromaDB, OpenCV, Scikit-learn, Pandas, Numpy, Matplotlib, Seaborn
Domains: LLMs, Generative AI, Computer Vision, Database Management, NLP, Data Science, DevOps